

NAHA TAHUNA, Indonesia

WMO: 970080

Lat: 3.6859N

Lon: 125.5280E

Elev: 38

StdP: 100.87

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.3	22.9	20.5	15.2	25.5	21.1	15.8	25.8	7.4	27.8	6.0	27.2	0.1	N/A	0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.9	31.9	25.7	31.5	25.8	31.1	25.8	27.1	30.1	26.7	29.8	26.6	29.7	3.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.7	29.0	25.9	21.3	28.8	25.7	21.1	28.6	85.3	30.3	84.4	30.2	83.6	30.0	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 6.3	5.3	4.4	DB	20.2	34.0	2.9	2.3	18.1	35.6	16.4	36.9	14.7	38.2	12.6	39.9	(4)
(5)			WB	19.5	28.0	2.4	0.7	17.8	28.5	16.4	28.9	15.0	29.3	13.2	29.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	26.8	26.8	27.1	27.3	27.5	27.3	27.0	27.3	27.1	27.3	27.1	27.2
	DBStd	0.79	0.80	0.91	0.74	0.74	0.67	0.78	0.75	0.89	0.68	0.71	0.69	0.72
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6258	521	470	529	519	543	518	527	535	513	535	514	533
	CDD18.3	3216	263	237	271	269	284	268	269	277	263	277	264	275
	CDH23.3	32656	2526	2288	2764	2833	3067	2709	2652	2881	2709	2827	2593	2807
(13) Wind	WSAvg	1.2	1.3	1.4	1.2	1.0	1.1	1.2	1.5	1.6	1.3	1.2	0.9	1.0
	PrecAvg	3355	451	307	275	225	264	260	212	162	161	232	362	444
	PrecMax	4421	865	592	726	437	624	407	394	382	372	441	698	798
	PrecMin	1768	142	80	23	42	71	5	14	10	4	34	87	72
	PrecStd	661	184	132	162	119	122	91	103	80	101	119	145	154
	DB	31.2	31.1	31.1	31.6	32.0	32.0	32.0	32.5	32.1	32.2	31.8	31.6	31.6
	MCWB	25.9	25.8	25.9	25.8	26.1	26.1	25.8	25.8	24.9	25.7	26.1	26.1	26.1
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	30.4	30.2	30.4	30.9	31.3	31.2	31.2	31.7	31.4	31.3	31.0	30.9	30.9
	2% MCWB	25.8	25.6	25.8	26.1	26.2	26.1	25.6	25.5	25.1	25.8	26.2	26.1	26.1
	5% DB	29.9	29.8	30.0	30.4	30.9	30.7	30.7	31.2	30.9	30.8	30.5	30.3	30.3
	5% MCWB	25.7	25.5	25.7	26.1	26.3	26.1	25.5	25.4	25.4	25.8	26.2	26.1	26.1
	10% DB	29.4	29.2	29.6	30.0	30.5	30.2	30.1	30.6	30.4	30.4	30.0	29.8	29.8
	10% MCWB	25.6	25.4	25.6	26.0	26.3	26.0	25.4	25.3	25.4	25.8	26.1	26.1	26.1
	WB	26.7	26.6	26.7	27.1	27.4	27.2	26.7	26.7	26.7	26.9	27.1	27.1	27.1
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	29.5	29.7	29.6	29.9	30.3	30.2	30.3	30.7	30.1	30.4	30.2	29.7	29.7
	2% WB	26.4	26.2	26.4	26.7	27.0	26.7	26.2	26.2	26.4	26.6	26.7	26.7	26.7
	2% MCDB	29.2	29.1	29.3	29.6	30.1	29.9	29.9	30.3	29.9	30.1	29.8	29.5	29.5
	5% WB	26.1	25.9	26.1	26.5	26.7	26.5	26.0	26.0	26.1	26.3	26.5	26.6	26.6
	5% MCDB	29.0	28.8	29.1	29.4	29.8	29.7	29.7	30.0	29.7	29.8	29.6	29.3	29.3
	10% WB	25.9	25.7	25.9	26.2	26.5	26.2	25.7	25.7	25.8	26.1	26.2	26.3	26.3
	10% MCDB	28.6	28.4	28.8	29.1	29.5	29.4	29.2	29.6	29.4	29.6	29.2	29.0	29.0
(35) Mean Daily Temperature Range	MDBR	5.0	4.9	5.1	5.3	5.9	5.8	6.3	6.8	6.9	6.6	5.8	5.2	5.2
	5% DB	6.1	5.7	5.6	6.0	6.7	6.6	7.1	7.6	7.7	7.2	6.5	6.0	6.0
	MCDBR	2.6	2.4	2.3	2.5	2.9	2.7	2.8	2.8	3.1	3.0	2.8	2.6	2.6
	5% WB	5.5	5.1	5.3	5.5	5.9	6.0	6.4	7.0	6.6	6.5	5.9	5.4	5.4
	MCWBR	2.6	2.4	2.4	2.5	2.8	2.9	2.8	2.9	3.1	3.0	2.8	2.6	2.6
(40) Clear-Sky Solar Irradiance	taub	0.419	0.420	0.414	0.410	0.400	0.403	0.413	0.431	0.433	0.455	0.409	0.413	0.413
	taud	2.526	2.505	2.536	2.535	2.562	2.552	2.519	2.453	2.432	2.365	2.520	2.526	2.526
	Ebn at Noon	906	910	911	897	885	870	866	866	880	866	907	905	905
	Edh at Noon	108	112	109	107	101	100	104	114	119	127	108	107	107
(44) All-Sky Solar Radiation	RadAvg	4.82	5.44	5.79	5.97	5.57	4.91	5.15	5.64	5.76	5.60	5.20	4.96	4.96
	RadStd	0.45	0.53	0.55	0.57	0.43	0.36	0.51	0.37	0.36	0.34	0.41	0.42	0.42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page